Acidity of alcohols

According to Bronsted theory, an acid is defined as a proton donor and the acid strength is the tendency to give up a proton. Alcohols are weakly acidic and their acidity is comparable with water. Except methanol, all other alcohols are weaker acid than water. The K_a value for water is 1.8×10^{-16} where as for alcohols, the K_a value in the order 10^{-18} to 10^{-16}.

Alcohols react with active metals such as sodium, aluminium etc… to form the corresponding alkoxides with the liberation of hydrogen gas and similar reaction to give alkoxide is not observed in the reaction of alcohol with NaOH.

$$2C_2H_5\text{-}OH + 2Na \rightarrow 2C_2H_5ONa + H_2 \uparrow$$

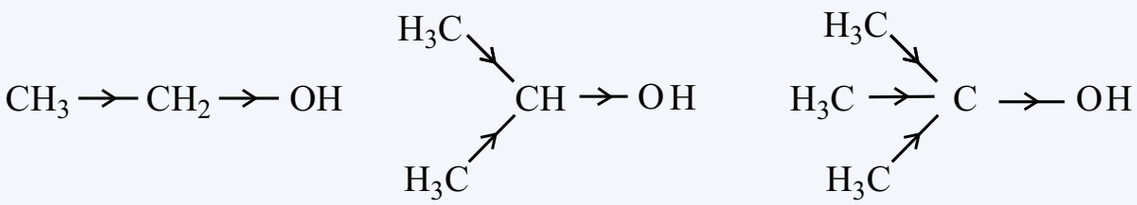
The above reaction explains the acidic nature of alcohols.

Comparison of acidity of 1°, 2°and 3° alcohols

The acidic nature of the alcohol is due to the polar nature of O –H bond. When an electron withdrawing -I groups such as -Cl, - F etc… is attached to the carbon bearing the OH group, it withdraws the electron density towards itself and thereby facilitating the proton donation. In contrast, the electron releasing group such as alkyl group increases the electron density on oxygen and decreases the polar nature of O – H bond, Hence it results in the decrease in acidity. on moving from primary to secondary and tertiary alcohols, the number of alkyl groups which attached to the carbon bearing -OH group increases, which results in the following order of acidity.

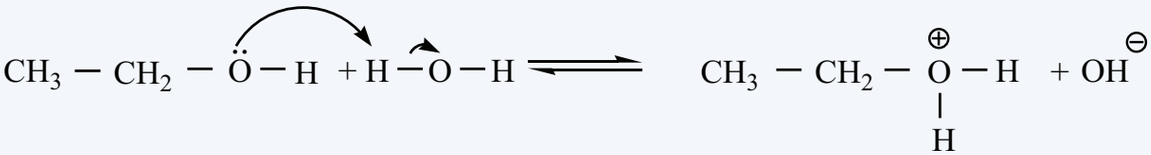
1° alcohol > 2° alcohol > 3° alcohol

For example



$K_a = 1.3\times10^{-16}$ $K_a = 3.2\times10^{-17}$ $K_a = 1\times10^{-18}$

Alcohols can also act as a Bronsted bases. It is due to the presence of unshared electron pairs on oxygen which make them proton acceptors.



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